

1. PRODUCT NAME

ZYTIGA® (abiraterone acetate) 250 mg tablets

ZYTIGA® (abiraterone acetate) 500 mg film-coated tablets

2. DOSAGE FORMS AND STRENGTHS

ZYTIGA tablet contains 250 mg of abiraterone acetate.

ZYTIGA film coated tablets contain 500 mg of abiraterone acetate.

3. PHARMACEUTICAL FORM

250 mg uncoated tablets

White to off-white oval tablets, debossed with “AA250” on one side.

500 mg film-coated tablets

Purple, oval shaped, film coated tablets, debossed with “AA” on one side and “500” on the other.

For excipients, see PHARMACEUTICAL INFORMATION – List of Excipients.

4. CLINICAL INFORMATION

4.1 Indications

ZYTIGA is indicated with prednisone or prednisolone for

- the treatment of metastatic castration resistant prostate cancer (mCRPC) in adult men who are asymptomatic or mildly symptomatic after failure of androgen deprivation therapy in whom chemotherapy is not yet clinically indicated (see section 5.1).
- the treatment of mCRPC in adult men whose disease has progressed on or after a docetaxel-based chemotherapy regimen.

ZYTIGA is also indicated in combination with prednisone or prednisolone and androgen deprivation therapy (ADT) for the treatment of patients with newly diagnosed high risk metastatic hormone sensitive prostate cancer (mHSPC) who may have received up to 3 months of prior ADT.

4.2 Posology and Method of Administration

Posology

The recommended dose is 1000 mg (two 500 mg tablets or four 250 mg tablets) as a single daily dose that must not be taken with food (see information on the method of administration). Taking the tablets with food increases systemic exposure to abiraterone (see sections 4.5 and 5.2).

Dosage of prednisone or prednisolone

For mCRPC, ZYTIGA is used with 10 mg prednisone or prednisolone daily.

For mHSPC, ZYTIGA is used with 5 mg prednisone or prednisolone daily.

Medical castration with LHRH analogue should be continued during treatment in patients not surgically castrated.

Recommended monitoring

Serum transaminases should be measured prior to starting treatment, every two weeks for the first three months of treatment and monthly thereafter. Blood pressure, serum potassium and fluid retention should be monitored monthly. However, patients with a significant risk or congestive heart failure

should be monitored every 2 weeks for the first three months of treatment and monthly thereafter (see section 4.4).

In patients with pre-existing hypokalaemia or those that develop hypokalaemia whilst being treated with ZYTIGA, consider maintaining the patient's potassium level at $\geq 4.0\text{mM}$.

For patients who develop Grade ≥ 3 toxicities including hypertension, hypokalaemia, oedema and other non-mineralocorticoid toxicities, treatment should be withheld and appropriate medical management should be instituted. Treatment with ZYTIGA should not be reinitiated until symptoms of the toxicity have resolved to Grade 1 or baseline.

In the event of a missed daily dose of either ZYTIGA, prednisone or prednisolone, treatment should be resumed the following day with the usual daily dose.

Hepatotoxicity

For patients who develop hepatotoxicity during treatment (alanine aminotransferase [ALT] increases or aspartate aminotransferase [AST] increases above 5 times the upper limit of normal [ULN]), treatment should be withheld immediately (see section 4.4). Re-treatment following return of liver function tests to the patient's baseline may be given at a reduced dose of 500 mg (one 500 mg tablet or two 250 mg tablets) once daily. For patients being re-treated, serum transaminases should be monitored at a minimum of every two weeks for three months and monthly thereafter. If hepatotoxicity recurs at the reduced dose of 500 mg daily, treatment should be discontinued.

If patients develop severe hepatotoxicity (ALT or AST 20 times the upper limit of normal) anytime while on therapy, treatment should be discontinued and patients should not be re-treated.

Hepatic impairment

No dose adjustment is necessary for patients with pre-existing mild hepatic impairment, Child-Pugh Class A.

Moderate hepatic impairment (Child-Pugh Class B) has been shown to increase the systemic exposure to abiraterone by approximately four-fold following single oral doses of abiraterone acetate 1000 mg (see section 5.2). There are no data on the clinical safety and efficacy of multiple doses of abiraterone acetate when administered to patients with moderate or severe hepatic impairment (Child-Pugh Class B or C). No dose adjustment can be predicted. The use of ZYTIGA should be cautiously assessed in patients with moderate hepatic impairment, in whom the benefit clearly should outweigh the possible risk. ZYTIGA should not be used in patients with severe hepatic impairment (see sections 4.3, 4.4 and 5.2).

Renal impairment

No dose adjustment is necessary for patients with renal impairment (see section 5.2). However, there is no clinical experience in patients with prostate cancer and severe renal impairment. Caution is advised in these patients (see section 4.4).

Paediatric population

There is no relevant use of this medicinal product in the paediatric population, as prostate cancer is not present in children and adolescents.

Method of administration

ZYTIGA is for oral use.

The tablets must be taken as a single dose once daily on an empty stomach. ZYTIGA must be taken at least two hours after eating and food must not be eaten for at least one hour after taking ZYTIGA. ZYTIGA tablets must be swallowed whole with water.

4.3 Contraindications

- Hypersensitivity to the active substance or to any of the excipients (see 6.1).

- Women who are or may potentially be pregnant (see 4.6).
- Severe hepatic impairment [(Child-Pugh Class C) (see 4.2, 4.4 and 5.2)].
- ZYTIGA with prednisone or prednisolone is contraindicated in combination with Ra-223.

4.4 Special Warnings and Precautions for use

Hypertension, hypokalaemia and fluid retention due to mineralocorticoid excess

ZYTIGA may cause hypertension, hypokalaemia and fluid retention (see 4.8) as a consequence of increased mineralocorticoid levels resulting from CYP17 inhibition (see 5.1-Mechanism of action). Co-administration of a corticosteroid suppresses adrenocorticotrophic hormone (ACTH) drive, resulting in a reduction in the incidence and severity of these adverse reactions. Caution is required in treating patients whose underlying medical conditions might be compromised by increases in blood pressure, hypokalaemia or fluid retention, e.g., those with heart failure, recent myocardial infarction or ventricular arrhythmia. In postmarketing experience, QT prolongation and Torsades de Pointes have been observed in patients who develop hypokalemia or have underlying cardiovascular conditions while taking ZYTIGA.

ZYTIGA should be used with caution in patients with a history of cardiovascular disease. The safety of ZYTIGA in patients with left ventricular ejection fraction (LVEF) <50% or New York Heart Association (NYHA) Class III or IV heart failure (in Study 301) or NYHA Class II to IV heart failure (in Studies 3011 and 302) was not established (see 4.8 and 5.1-Clinical Studies). Before treatment with ZYTIGA, hypertension must be controlled and hypokalemia must be corrected. Blood pressure, serum potassium and fluid retention should be monitored at least monthly.

Hepatotoxicity and hepatic impairment

Marked increases in liver enzymes leading to drug discontinuation or dosage modification occurred in controlled clinical studies (see 4.8). Serum transaminase and bilirubin levels should be measured prior to starting treatment with ZYTIGA, every two weeks for the first three months of treatment, and monthly thereafter. If clinical symptoms or signs suggestive of hepatotoxicity develop, serum transaminases should be measured immediately. If at any time the ALT or AST rises above 5 times the upper limit of normal or the bilirubin rises above 3 times the upper limit of normal, treatment with ZYTIGA should be interrupted immediately and liver function closely monitored.

Re-treatment with ZYTIGA may only take place after the return of liver function tests to the patient's baseline and at a reduced dose level (see 4.2).

If patients develop severe hepatotoxicity (ALT or AST 20 times the upper limit of normal) anytime while on therapy, ZYTIGA should be discontinued and patients should not be re-treated with ZYTIGA.

There are no data on the clinical safety and efficacy of multiple doses of abiraterone acetate when administered to patients with moderate or severe hepatic impairment (Child Pugh Class B or C). No dose adjustment can be predicted. ZYTIGA should be used with caution in patients with moderate hepatic impairment only if the benefit clearly outweighs the possible risk (see 4.2 and 5.2–Special populations). ZYTIGA should not be used in patients with severe hepatic impairment (see 4.2 and 5.2–Special populations).

There have been rare post-marketing reports of acute liver failure and hepatitis fulminant, some with fatal outcome (see 4.8).

Corticosteroid withdrawal and coverage of stress situations

Caution is advised and monitoring for adrenocortical insufficiency should occur if patients need to be withdrawn from prednisone or prednisolone. If ZYTIGA is continued after corticosteroids are withdrawn, patients should be monitored for symptoms of mineralocorticoid excess (see 4.4-Hypertension, hypokalemia and fluid retention due to mineralocorticoid excess).

In patients on prednisone or prednisolone who are subjected to unusual stress, increased dosage of a corticosteroid may be indicated before, during and after the stressful situation.

Hypoglycemia

Cases of hypoglycemia have been reported when ZYTIGA was administered to patients with pre-existing diabetes receiving pioglitazone or repaglinide; therefore, blood sugar should be measured frequently in patients with diabetes.

Use with chemotherapy

The safety and efficacy of concomitant use of ZYTIGA with cytotoxic chemotherapy has not been established (see 5.1-Clinical studies).

Combination of abiraterone and prednisone/prednisolone with Ra-223

Treatment with abiraterone and prednisone/prednisolone in combination with Ra-223 is contraindicated (see 4.3) due to an increased risk of fractures and a trend for increased mortality among asymptomatic or mildly symptomatic prostate cancer patients as observed in clinical trials.

It is recommended that subsequent treatment with Ra-223 is not initiated for at least 5 days after the last administration of ZYTIGA in combination with prednisone/prednisolone.

Bone density

Decreased bone density may occur in men with metastatic advanced prostate cancer (castration resistant prostate cancer). The use of ZYTIGA in combination with a glucocorticoid could increase this effect.

Prior use of ketoconazole

Lower rates of response might be expected in patients previously treated with ketoconazole for prostate cancer.

Intolerance to excipients

This medicinal product contains lactose. Patients with rare hereditary problems of galactose intolerance, total lactase deficiency or glucose-galactose malabsorption should not take this medicine. This medicinal product contains 27.2 mg (1.18 mmol) sodium per dose of four tablets, equivalent to 1.36% of the WHO recommended maximum daily intake of 2 g sodium for an adult.

4.5 Interactions

Effect of food on abiraterone

Administration of ZYTIGA with food significantly increases the absorption of abiraterone. The efficacy and safety of ZYTIGA given with food has not been established. **ZYTIGA must not be taken with food** (see 4.2 and 5.2-Absorption).

Interactions with other drugs

Potential for other drugs to affect abiraterone exposures

In a clinical pharmacokinetic interaction study of healthy subjects pretreated with a strong CYP3A4 inducer (rifampicin, 600 mg daily for 6 days) followed by a single dose of abiraterone acetate 1,000 mg, the mean plasma AUC_∞ of abiraterone was decreased by 55%.

Strong inducers of CYP3A4 (e.g., phenytoin, carbamazepine, rifampicin, rifabutin, rifapentine, phenobarbital) during treatment with ZYTIGA are to be avoided, or used with careful evaluation of clinical efficacy.

In a separate clinical pharmacokinetic interaction study of healthy subjects, co-administration of ketoconazole, a strong inhibitor of CYP3A4, had no clinically meaningful effect on the pharmacokinetics of abiraterone.

Potential for ZYTIGA to affect exposures to other drugs

Abiraterone is an inhibitor of the hepatic drug-metabolizing enzymes CYP2D6 and CYP2C8. In a clinical study to determine the effects of abiraterone acetate (plus prednisone) on a single dose of the CYP2D6 substrate dextromethorphan, the systemic exposure (AUC) of dextromethorphan was increased by approximately 200%. The AUC₂₄ for dextrophan, the active metabolite of dextromethorphan, increased approximately 33%.

Caution is advised when ZYTIGA is administered with drugs activated by or metabolized by CYP2D6, particularly with drugs that have a narrow therapeutic index. Dose reduction of narrow therapeutic index drugs metabolized by CYP2D6 should be considered.

In the same study to determine the effects of abiraterone acetate (plus prednisone) on a single dose of the CYP1A2 substrate theophylline, no increase in systemic exposure of theophylline was observed.

In a CYP2C8 drug-drug interaction trial in healthy subjects, the AUC of pioglitazone was increased by 46% and the AUCs for M-III and M-IV, the active metabolites of pioglitazone, each decreased by 10% when pioglitazone was given together with a single dose of 1000 mg abiraterone acetate. Patients should be monitored for signs of toxicity related to a CYP2C8 substrate with a narrow therapeutic index if used concomitantly with ZYTIGA. Examples of medicinal products metabolized by CYP2C8 include pioglitazone and repaglinide.

4.6 Pregnancy, Breast-feeding and Fertility

Pregnancy

ZYTIGA is contraindicated in women who are or may potentially be pregnant (see 4.3).

There are no human data on the use of ZYTIGA in pregnancy and ZYTIGA is not for use in women of child-bearing potential. Maternal use of a CYP17 inhibitor is expected to produce changes in hormone levels that could affect development of the fetus (see 5.1-Mechanism of action and 5.3-Reproductive Toxicology).

It is not known if abiraterone or its metabolites are present in semen. A condom is required if the patient is engaged in sexual activity with a pregnant woman. If the patient is engaged in sex with a woman of child-bearing potential, a condom is required along with another effective contraceptive method.

To avoid inadvertent exposure, women who are pregnant or women who may be pregnant should not handle ZYTIGA 250 mg uncoated tablets without protection, e.g., gloves.

Breast-feeding

ZYTIGA is not for use in women.

It is not known if either abiraterone or its metabolites are excreted in human breast milk.

4.7 Effects on Ability to Drive and Use Machines

No studies on the effects of ZYTIGA on the ability to drive or use machines have been performed. It is not anticipated that ZYTIGA will affect the ability to drive and use machines.

4.8 Adverse Reactions

Throughout this section, adverse reactions are presented. Adverse reactions are adverse events that were considered to be reasonably associated with the use of abiraterone acetate based on the comprehensive assessment of the available adverse event information. A causal relationship with abiraterone acetate cannot be reliably established in individual cases. Further, because clinical trials are conducted under widely varying conditions, adverse reaction rates observed in the clinical trials of a drug cannot be directly compared to rates in the clinical trials of another drug and may not reflect the rates observed in clinical practice.

In an analysis of adverse reaction of composite Phase 3 studies with ZYTIGA, adverse reactions that were observed in $\geq 10\%$ of patients were hypertension, peripheral edema, hypokalemia, urinary tract infection, and aspartate aminotransferase increased and/or alanine aminotransferase increased.

ZYTIGA may cause hypertension, hypokalemia and fluid retention as a pharmacodynamic consequence of its mechanism of action. In Phase 3 studies anticipated mineralocorticoid effects were seen more commonly in patients treated with ZYTIGA versus patients treated with placebo: hypokalemia 18% versus 8%, hypertension 22% versus 16% and fluid retention (peripheral edema) 23% versus 17%, respectively. In patients treated with ZYTIGA, grades 3 and 4 hypokalemia were observed in 6% and 2% of patients, grades 3 and 4 hypertension were observed in 8% and 5% of patients, and grades 3 and 4 fluid retention edema were observed in 1% and 1% of patients, respectively. Mineralocorticoid effects generally were able to be successfully managed medically. Concomitant use of a corticosteroid reduces the incidence and severity of these adverse reactions (see 4.4-Hypertension, hypokalemia and fluid retention due to mineralocorticoid excess).

In studies of patients with metastatic advanced prostate cancer who were using a LHRH agonist, or were previously treated with orchiectomy, ZYTIGA was administered at a dose of 1000 mg daily in combination with low dose prednisone or prednisolone (5 or 10 mg daily).

Adverse reactions that occurred at a rate of $\geq 1\%$ (all grades) are shown in Table 1a:

Table 1a: Adverse Reactions Due to ZYTIGA in $\geq 1\%$ of Patients in a Clinical Studies^a

	ZYTIGA 1000 mg daily with prednisone or prednisolone n = 2659 ^b		
System Organ Class Adverse Reaction	All grades %	Grade 3 %	Grade 4 %
General Disorders and Administration Site Conditions			
Edema peripheral	20	<1	0
Metabolism and Nutrition Disorders			
Hypokalemia	20	5	<1
Hypertriglyceridemia	1	0	0
Infections and Infestations			
Urinary tract infection	10	2	<1
Hepatobiliary Disorders			
ALT increased and/or AST increased ^c	13	4	<1
Vascular Disorders			
Hypertension	21	6	0
Injury, Poisoning and Procedural Complications			
Fractures ^d	7	2	<1
Cardiac Disorders			
Cardiac failure ^c	1	<1	<1
Angina pectoris	2	<1	0
Arrhythmia	1	0	0
Atrial fibrillation	3	1	<1
Tachycardia	2	<1	0
Renal and Urinary Disorders			
Hematuria	7	1	0
Gastrointestinal Disorders			
Dyspepsia	6	0	0

^a All patients were using an LHRH agonist or had undergone orchiectomy.

^b n = patients assessed for safety.

^c ALT increased and/or AST increased includes ALT increased, AST increased, and hepatic function abnormal.

^d Fractures includes all fractures with the exception of pathological fractures.

^e Cardiac failure also includes congestive heart failure, left ventricular dysfunction and ejection fraction decreased.

The adverse reaction, adrenal insufficiency, occurred in the Phase 3 clinical studies at a rate of 0.3% in patients taking ZYTIGA and at a rate of 0.1% in patients taking placebo.

Cardiovascular effects

The three Phase 3 studies excluded patients with uncontrolled hypertension, clinically significant heart disease as evidenced by myocardial infarction, arterial thrombotic events in the past 6 months, severe or unstable angina, NYHA Class III or IV heart failure (Study 301) or Class II to IV heart failure (Studies 3011 and 302) or cardiac ejection fraction measurement of < 50%. All patients enrolled (both active and placebo-treated patients) were concomitantly treated with androgen deprivation therapy, predominantly with the use of LHRH agonists, which has been associated with diabetes, myocardial infarction, cerebrovascular accident and sudden cardiac death. The incidence of cardiovascular adverse reactions in the Phase 3 studies in patients taking ZYTIGA versus patients taking placebo were as follows: atrial fibrillation 2.6% vs. 2.0%, tachycardia 1.9% vs. 1.0%, angina pectoris 1.7% vs. 0.8%, cardiac failure 0.7% vs. 0.2% and arrhythmia 0.7% vs. 0.5%.

Hepatotoxicity

Drug associated hepatotoxicity with elevated ALT, aspartate transaminase (AST) and total bilirubin has been reported in patients treated with ZYTIGA. Across Phase 3 clinical studies, hepatotoxicity grades 3 and 4 (e.g., ALT or AST increases of > 5X ULN or bilirubin increases > 1.5XULN) were reported in approximately 6% of patients who received ZYTIGA, typically during the first 3 months after starting treatment. In Study 3011, grade 3 or 4 hepatotoxicity was observed in 8.4% of patients treated with ZYTIGA. Ten patients who received ZYTIGA were discontinued because of hepatotoxicity; two had Grade 2 hepatotoxicity, six had Grade 3 hepatotoxicity, and two had Grade 4 hepatotoxicity. No patient died of hepatotoxicity in Study 3011. In the Phase 3 clinical studies, patients whose baseline ALT or AST was elevated were more likely to experience liver function test elevations than those beginning with normal values. When elevations of either ALT or AST > 5X ULN, or elevations in bilirubin >3X ULN were observed, ZYTIGA was withheld or discontinued. In two instances marked increases in liver function tests occurred (see 4.4-Hepatotoxicity and Hepatic impairment). These two patients with normal baseline hepatic function, experienced ALT or AST elevations 15 to 40 XULN and bilirubin elevations 2 to 6X ULN. Upon discontinuation of ZYTIGA, both patients had normalisation of their liver function tests and one patient was re-treated with ZYTIGA without recurrence of the elevations. In Study 302, grade 3 or 4 ALT or AST elevations were observed in 35 (6.5%) patients treated with ZYTIGA. Aminotransferase elevations resolved in all but 3 patients (2 with new multiple liver metastases and 1 with AST elevation approximately 3 weeks after the last dose of ZYTIGA). In Phase 3 clinical studies, treatment discontinuations due to ALT and AST increases or abnormal hepatic function were reported in 1.1% of patients treated with ZYTIGA and 0.6% of patients treated with placebo; no deaths were reported due to hepatotoxicity event.

In clinical trials, the risk for hepatotoxicity was mitigated by exclusion of patients with baseline hepatitis or significant abnormalities of liver function tests. In the 3011 trial, patients with baseline ALT and AST > 2.5X ULN, bilirubin >1.5X ULN or those with active or symptomatic viral hepatitis or chronic liver disease; ascites or bleeding disorders secondary to hepatic dysfunction were excluded. In the 301 trial, patients with baseline ALT and AST $\geq$ 2.5X ULN in the absence of liver metastases and > 5X ULN in the presence of liver metastases were excluded. In the 302 trial patients with liver metastases were not eligible and patients with baseline ALT and AST $\geq$ 2.5x ULN were excluded. Abnormal liver function tests developing in patients participating in clinical trials were vigorously managed by requiring treatment interruption and permitting re-treatment only after return of liver function tests to the patient's baseline (see 4.2). Patients with elevations of ALT or AST > 20X ULN were not re-treated. The safety of re-treatment in such patients is unknown. The mechanism for hepatotoxicity associated with ZYTIGA is not understood.

Post-marketing experience

Adverse reactions identified during the post-marketing experience based on spontaneous reports with ZYTIGA are described below. The frequencies are provided according to the following convention:

Uncommon $\geq 1/1000$ and $< 1/100$, Rare $\geq 1/10000$ and $< 1/1000$, Very Rare $< 1/10000$

System Organ Class: Respiratory, thoracic and mediastinal disorders

Rare: Allergic alveolitis

System Organ Class: Musculoskeletal and connective tissue disorders

Uncommon: Rhabdomyolysis, Myopathy

System Organ Class: Hepatobiliary disorders

Rare: Hepatitis fulminant, Acute hepatic failure

System Organ Class: Cardiac disorders

Very rare: QT prolongation and Torsades de Pointes (observed in patients who developed hypokalemia or had underlying cardiovascular conditions).

System Organ Class: Immune System Disorders – Hypersensitivity

Very rare: Anaphylactic reaction (severe allergic reactions that include, but are not limited to difficulty swallowing or breathing, swollen face, lips, tongue or throat, or an itchy rash (urticaria)).

4.9 Overdose

Human experience of overdose with ZYTIGA is limited.

There is no specific antidote. In the event of an overdose, administration of ZYTIGA should be stopped and general supportive measures undertaken, including monitoring for arrhythmias. Liver function also should be assessed.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic Properties

Pharmacotherapeutic group: Other hormone antagonists and related agents, **ATC code:** L02BX03

Mechanism of action

Abiraterone acetate (ZYTIGA) is converted *in vivo* to abiraterone, an androgen biosynthesis inhibitor. Specifically, abiraterone selectively inhibits the enzyme 17 α -hydroxylase/C17,20-lyase (CYP17). This enzyme is expressed in and is required for androgen biosynthesis in testicular, adrenal and prostatic tumor tissues. It catalyzes the conversion of pregnenolone and progesterone into testosterone precursors, DHEA and androstenedione, respectively, by 17 α -hydroxylation and cleavage of the C17,20 bond. CYP17 inhibition also results in increased mineralocorticoid production by the adrenals (see 4.4-Hypertension, hypokalemia and fluid retention due to mineralocorticoid excess).

Androgen-sensitive prostatic carcinoma responds to treatment that decreases androgen levels. Androgen deprivation therapies, such as treatment with LHRH agonists or orchiectomy, decrease androgen production in the testes but do not affect androgen production by the adrenals or in the tumor. Treatment with ZYTIGA decreases serum testosterone to undetectable levels (using commercial assays) when given with LHRH agonists (or orchiectomy).

Pharmacodynamic effects

ZYTIGA decreases serum testosterone and other androgens to levels lower than those achieved by the use of LHRH agonists alone or by orchiectomy. This results from the selective inhibition of the CYP17 enzyme required for androgen biosynthesis. Prostate specific antigen (PSA) serves as a biomarker in patients with prostate cancer. In a Phase 3 clinical study of patients who failed prior chemotherapy with taxanes, 38% of patients treated with ZYTIGA, versus 10% of patients treated with placebo, had at least a 50% decline from baseline in PSA levels.

Use of Spironolactone

Patients in pivotal clinical trials with ZYTIGA were not allowed to use spironolactone as spironolactone binds to the androgen receptor and may increase PSA levels.

Clinical studies

The efficacy of ZYTIGA was established in three randomized placebo-controlled multicenter Phase 3 clinical studies (Studies 3011, 302, and 301) of patients with mHSPC and mCRPC.

Study 3011 enrolled patients who were newly diagnosed (within 3 months of randomization) mHSPC who had high-risk prognostic factors. High-risk prognosis was defined as having at least 2 of the following 3 risk factors: (1) Gleason score of ≥ 8 ; (2) presence of 3 or more lesions on bone scan; (3) presence of measurable visceral (excluding lymph node disease) metastasis. In the active arm, ZYTIGA was administered at a dose of 1000 mg daily in combination with low dose prednisone or prednisolone 5 mg once daily in addition to ADT (LHRH agonist or orchiectomy), which was the standard of care treatment. Patients in the control arm received ADT and placebos for both ZYTIGA and prednisone.

Study 302 enrolled patients who were asymptomatic or mildly symptomatic and had not received prior chemotherapy, whereas Study 301 enrolled patients who received prior chemotherapy containing a taxane. In both studies patients were using an LHRH agonist or were previously treated with orchiectomy. In the active treatment arms, ZYTIGA was administered at a dose of 1000 mg daily in combination with low dose prednisone or prednisolone 5 mg twice daily. Control patients received placebo and low dose prednisone or prednisolone 5 mg twice daily.

Because changes in PSA serum concentration do not always predict clinical benefit, in all studies patients were maintained on ZYTIGA until discontinuation criteria were met as specified for each study below.

Study 3011 (patients with newly diagnosed high risk mHSPC)

In Study 3011, (n=1199) the median age of enrolled patients was 67 years. The number of patients treated with ZYTIGA by racial group was Caucasian 832 (69.4%), Asian 246 (20.5%), Black or African American 25 (2.1%), other 80 (6.7%), unknown/not reported 13 (1.1%), and American Indian or Alaska Native 3 (0.3%). The ECOG performance status was 0 or 1 for 97% of patients. Patients with known brain metastasis, uncontrolled hypertension, significant heart disease, or NYHA Class II-IV heart failure were excluded. Patients that were treated with prior pharmacotherapy, radiation therapy, or surgery for metastatic prostate cancer were excluded with the exception of up to 3 months of ADT or 1 course of palliative radiation or surgical therapy to treat symptoms resulting from metastatic disease. Co-primary efficacy endpoints were overall survival (OS) and radiographic progression-free survival (rPFS). The median baseline pain score, as measured by the Brief Pain Inventory Short Form (BPI-SF) was 2.0 in both the treatment and Placebo groups. In addition to the co-primary endpoint measures, benefit was also assessed using time to skeletal-related event (SRE), time to subsequent therapy for prostate cancer, time to initiation of chemotherapy, time to pain progression, and time to PSA progression. Treatment continued until disease progression, withdrawal of consent, the occurrence of unacceptable toxicity, or death.

Radiographic progression-free survival was defined as the time from randomisation to the occurrence of radiographic progression or death from any cause. Radiographic progression included progression by bone scan (according to modified PCWG2) or progression of soft tissue lesions by CT or MRI (according to RECIST 1.1).

A significant difference in rPFS between treatment groups was observed (see Table 2 and Figure 1).

Table 2: Radiographic Progression-Free Survival - Stratified Analysis; Intent-to-treat Population (Study PCR3011)

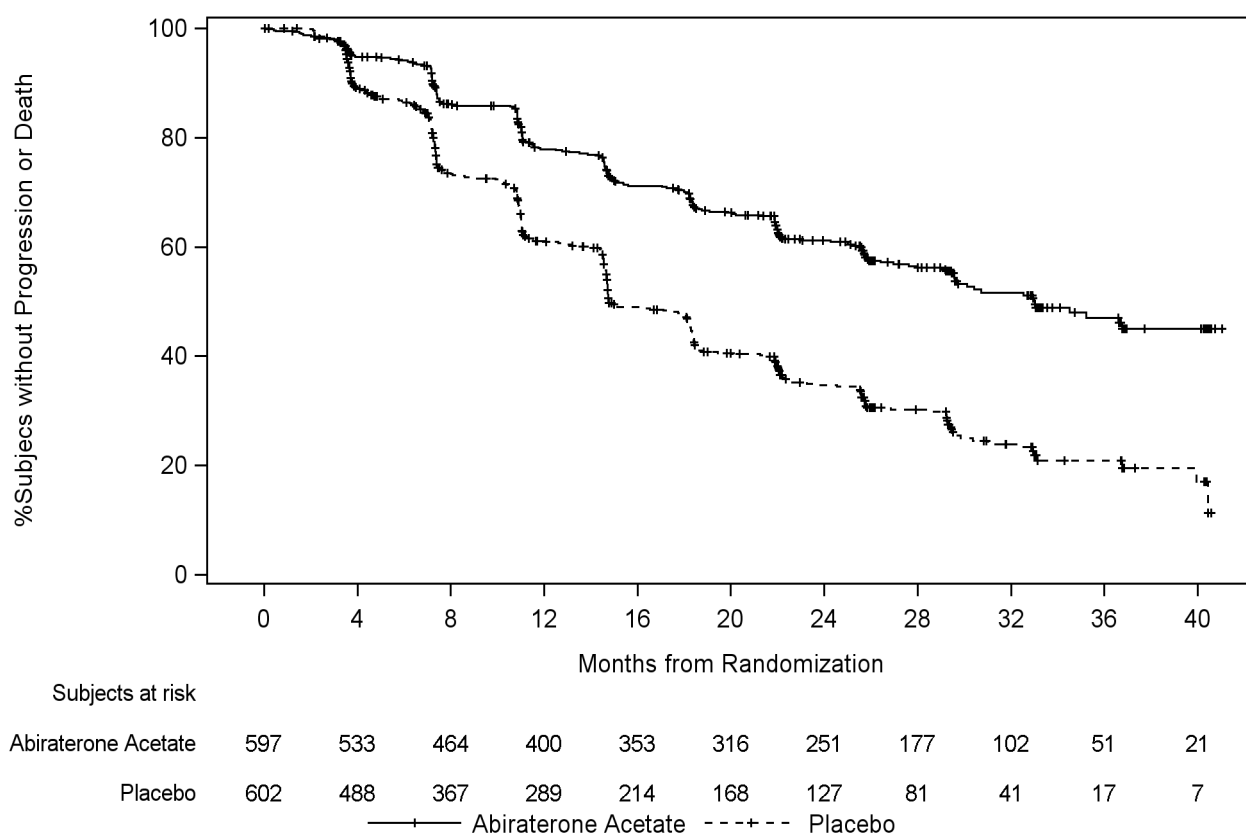
	AA-P	Placebo
Subjects randomised	597	602
Event	239 (40.0%)	354 (58.8%)
Censored	358 (60.0%)	248 (41.2%)
Time to Event (months)		
Median (95% CI)	33.02 (29.57, NE)	14.78 (14.69, 18.27)
Range	(0.0+, 41.0+)	(0.0+, 40.6+)
p value ^a	< 0.0001	
Hazard ratio (95% CI) ^b	0.466 (0.394, 0.550)	

Note: += censored observation, NE=not estimable. The radiographic progression and death are considered in defining the rPFS event. AA-P= subjects who received abiraterone acetate and prednisone.

^a p value is from a log-rank test stratified by ECOG PS score (0/1 or 2) and visceral lesion (absent or present).

^b Hazard ratio is from stratified proportional hazards model. Hazard ratio <1 favours AA-P.

Figure 1: Kaplan-Meier Plot of Radiographic Progression-free Survival; Intent-to-treat Population (Study PCR3011)



A statistically significant improvement in OS in favour of AA-P plus ADT was observed with a 38% reduction in the risk of death compared to Placebo plus ADT (HR=0.621; 95% CI: 0.509, 0.756; p<0.0001), crossing the pre-specified boundary for OS at Interim Analysis 1 of 0.010 (see Table 3 and Figure 2).

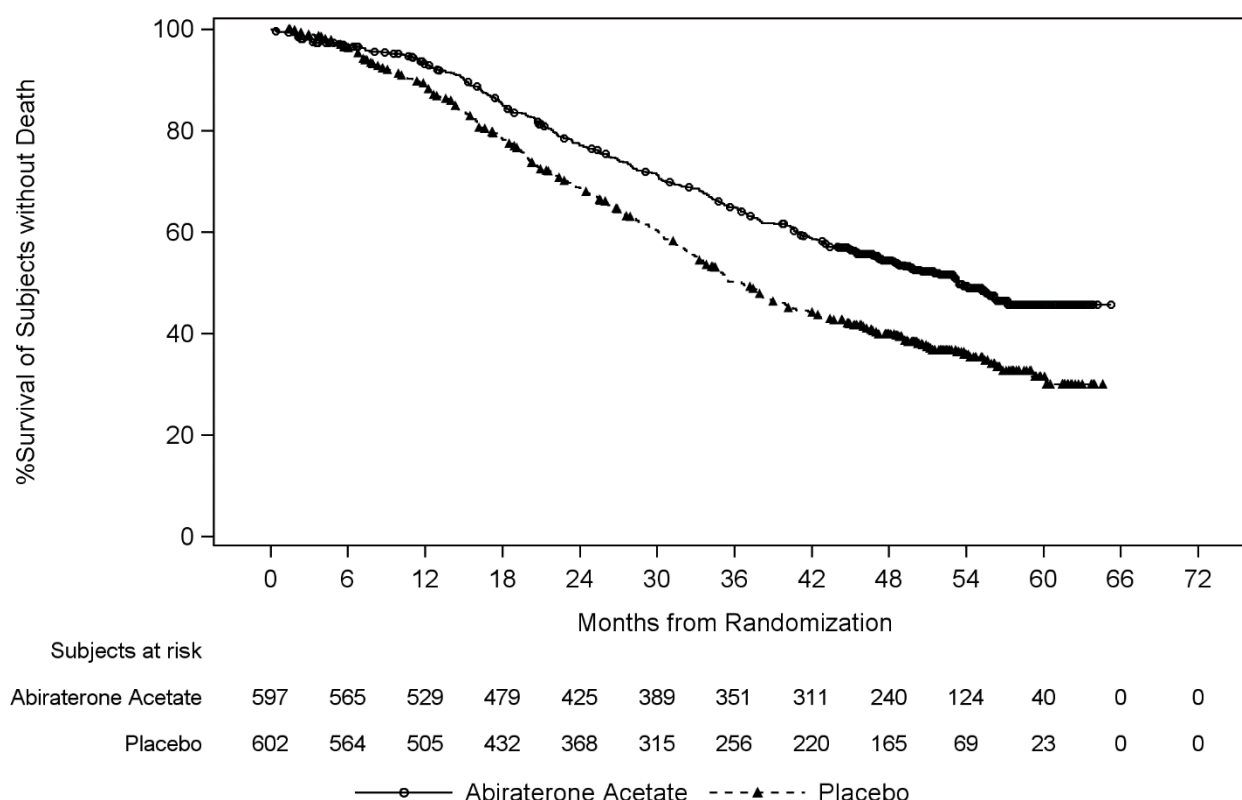
Table 3: Overall Survival of Patients Treated with Either ZYTIGA or Placebos in Study PCR3011 (Intent-to-Treat Analysis)

Overall Survival	ZYTIGA with Prednisone (N=597)	Placebos (N=602)
Deaths (%)	275 (46%)	343 (57%)
Median survival (months) (95% CI)	53.3 (48.2, NE)	36.5 (33.5, 40.0)
Hazard ratio (95% CI) ¹	0.66 (0.56, 0.78)	

NE=Not estimable

¹ Hazard Ratio is derived from a stratified proportional hazards model. Hazard ratio <1 favours ZYTIGA with prednisone.

Figure 2: Kaplan-Meier Plot of Overall Survival; Intent-to-treat Population in Study PCR3011 Analysis



Subgroup analyses consistently favour treatment with ZYTIGA. The treatment effect of AA-P on rPFS and OS across the pre-specified subgroups was favourable and consistent with the overall study population, except for the subgroup of ECOG score of 2 where no trend towards benefit was observed, however the small sample size (n=40) limits drawing any meaningful conclusion.

In addition to the observed improvements in overall survival and rPFS, benefit was demonstrated for ZYTIGA vs. placebo treatment in all prospectively-defined secondary endpoints.

Study 302 (asymptomatic or mildly symptomatic patients who did not receive prior chemotherapy)

In Study 302, (n=1088) the median age of enrolled patients was 71 years for patients treated with ZYTIGA plus prednisone or prednisolone and 70 years for patients treated with placebo plus prednisone or prednisolone. The ECOG performance status was 0 for 76% of patients, and 1 for 24% of patients in both arms. Patients with visceral metastases were excluded. Co-primary efficacy endpoints were overall survival and radiographic progression-free survival (rPFS). Baseline pain assessment was 0-1 (asymptomatic) in 66% of patients and 2-3 (mildly symptomatic) in 26% of patients as defined by the Brief Pain Inventory-Short Form (worst pain over the last 24 hours). In

addition to the co-primary endpoint measures, benefit was also assessed using time to opiate use for cancer pain, time to initiation of cytotoxic chemotherapy, time to deterioration in ECOG performance score by ≥ 1 point and time to PSA progression based on Prostate Cancer Working Group-2 (PCWG2) criteria.

In the 302 study treatments were discontinued at the time of unequivocal clinical progression. Treatments could also be discontinued at the time of confirmed radiographic progression at the discretion of the investigator.

Radiographic progression free survival was assessed with the use of sequential imaging studies as defined by PCWG2 criteria (for bone lesions) and modified Response Evaluation Criteria In Solid Tumours (RECIST) criteria (for soft tissue lesions). Analysis of rPFS utilized centrally-reviewed radiographic assessment of progression.

At the planned rPFS analysis there were 401 events; 150 (28%) of patients treated with ZYTIGA and 251 (46%) of patients treated with placebo had radiographic evidence of progression or had died.¹ A significant difference in rPFS between treatment groups was observed (see Table 4 and Figure 3).

Table 4: Study 302: Radiographic Progression-free Survival of Patients Treated with Either ZYTIGA or Placebo in Combination with Prednisone or Prednisolone Plus LHRH Agonists or Prior Orchiectomy

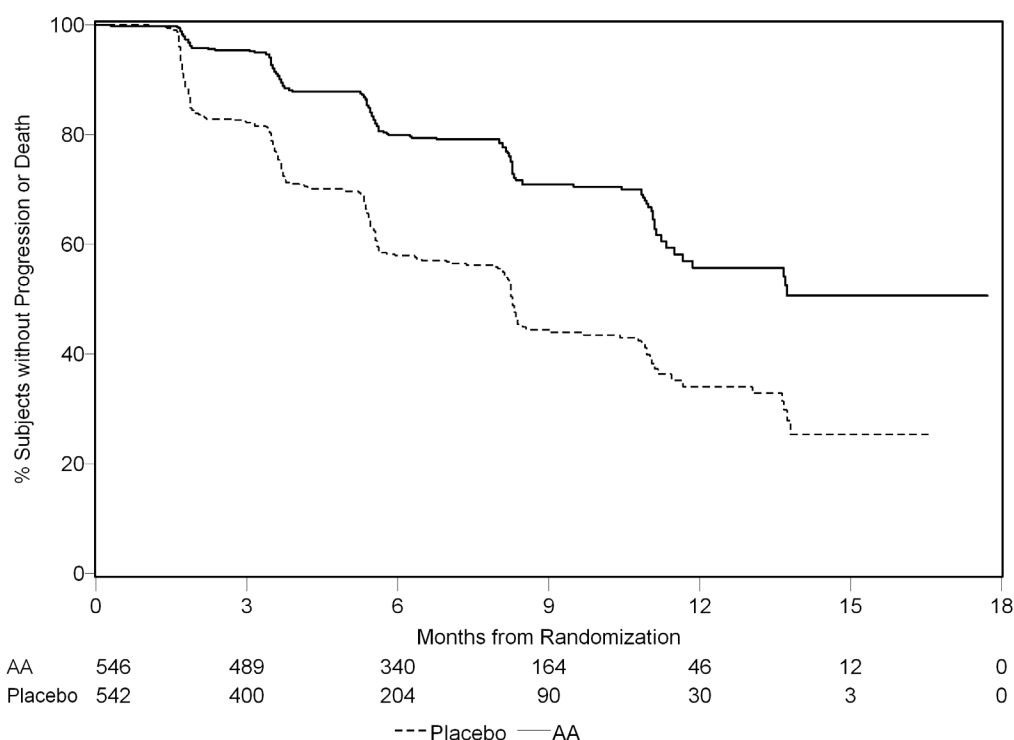
	ZYTIGA (N = 546)	PLACEBO (N = 542)
Radiographic Progression-free Survival (rPFS)		
Progression or death	150 (28%)	251 (46%)
Median rPFS in months (95% CI)	Not reached (11.66, NE)	8.3 (8.12, 8.54)
p value*	< 0.0001	
Hazard ratio** (95% CI)	0.425 (0.347, 0.522)	

NE = Not Estimated

* p-value is derived from a log-rank test stratified by baseline ECOG score (0 or 1)

** Hazard ratio (HR) < 1 favors ZYTIGA

Figure 3: Kaplan Meier Curves of Radiographic Progression-free Survival in Patients Treated with Either ZYTIGA or Placebo in Combination with Prednisone or Prednisolone plus LHRH Agonists or Prior Orchiectomy



AA = ZYTIGA

However, subject data continued to be collected through the date of the second interim analysis of Overall survival (OS). The investigator radiographic review of rPFS performed as a follow up sensitivity analysis is presented in Table 5 and Figure 4.

Six hundred and seven (607) subjects had radiographic progression or died: 271 (50%) in the abiraterone acetate group and 336 (62%) in the placebo group. Treatment with abiraterone acetate decreased the risk of radiographic progression or death by 47% compared with placebo (HR = 0.530; 95% CI: [0.451, 0.623], $p < 0.0001$). The median rPFS was 16.5 months in the abiraterone acetate group and 8.3 months in the placebo group.

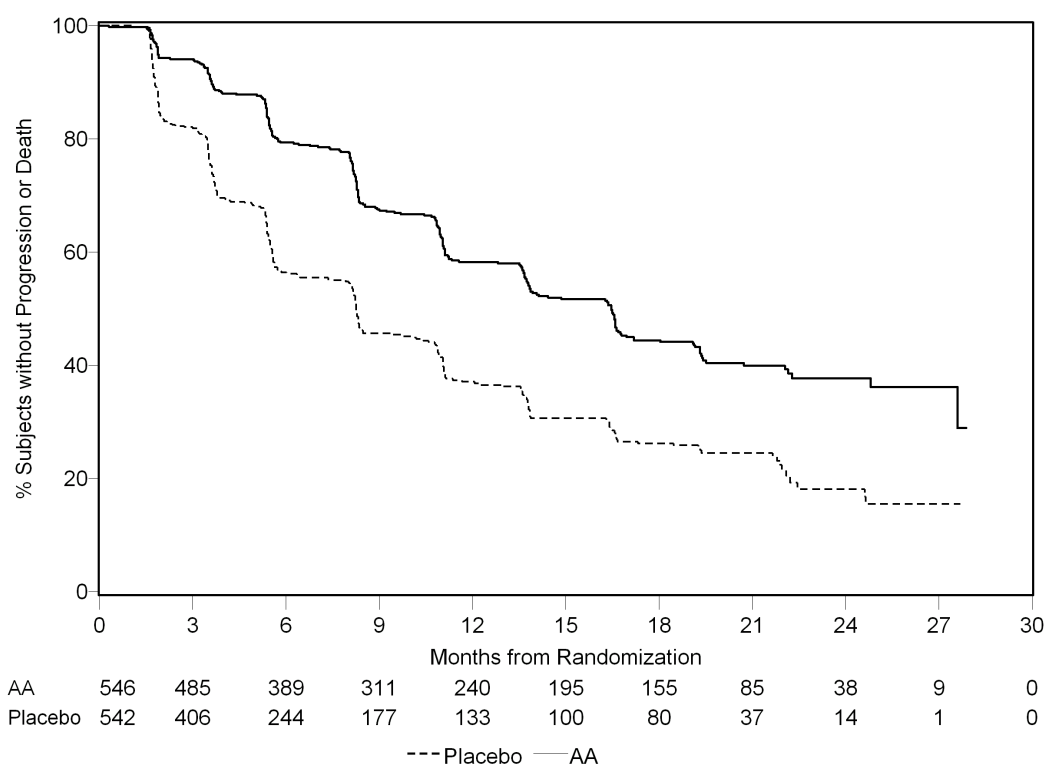
Table 5: Study 302: Radiographic Progression-free Survival Of Patients Treated With Either ZYTIGA Or Placebo In Combination With Prednisone Or Prednisolone Plus LHRH Analogues Or Prior Orchiectomy (At Second Interim Analysis Of OS-Investigator Review)

	ZYTIGA (N = 546)	PLACEBO (N = 542)
Radiographic Progression-free Survival (rPFS)		
Progression or death	271 (50%)	336 (62%)
Median rPFS in months (95% CI)	16.5 (13.80, 16.79)	8.3 (8.05, 9.43)
p value*	< 0.0001	
Hazard ratio** (95% CI)	0.530 (0.451, 0.623)	

* p-value is derived from a log-rank test stratified by baseline ECOG score (0 or 1)

** Hazard ratio < 1 favours ZYTIGA

Figure 4: Kaplan Meier Curves of Radiographic Progression Free Survival in Patients Treated with Either ZYTIGA or Placebo in Combination with Prednisone or Prednisolone plus LHRH Analogues or Prior Orchiectomy (At second interim analysis of OS Investigator Review)



AA = ZYTIGA

A planned interim analysis (IA) for overall survival was conducted after 333 deaths were observed. The study was unblinded based on the magnitude of clinical benefit observed and patients in the placebo group were offered treatment with ZYTIGA. Overall survival was longer for ZYTIGA than placebo with a 25% reduction in risk of death (H R = 0.752; 95 % CI: [0.606 , 0.934], p = 0.0097), but OS was not mature and interim results did not meet the pre-specified stopping boundary for statistical significance (see Table 6). Survival continued to be followed after this IA.

The planned final analysis for OS was conducted after 741 deaths were observed (median follow-up of 49 months). Sixty five percent (354 of 546) of patients treated with ZYTIGA, compared with 71% (387 of 542) of patients treated with placebo, had died. A statistically significant OS benefit in favor of the ZYTIGA-treated group was demonstrated with a 19.4% reduction in risk of death (HR = 0.806; 95% CI: [0.697, 0.931], p = 0.0033) and an improvement in median OS of 4.4 months (ZYTIGA 34.7 months, placebo 30.3 months) (see Table 6 and Figure 5). This improvement was demonstrated despite subsequent therapy being common, irrespective of whether patients initially received abiraterone acetate or placebo. Subsequent therapies in the abiraterone acetate and placebo patient groups included abiraterone acetate, 69 (13%) and 238 (44%); docetaxel, 311 (57%) and 331 (61%); cabazitaxel, 100 (18%) and 105 (19%); and enzalutamide 87 (16%) and 54 (10%) patients respectively.

Table 6: Study 302: Overall Survival of Patients Treated with Either ZYTIGA or Placebo in Combination with Prednisone or Prednisolone Plus LHRH Agonists or Prior Orchiectomy

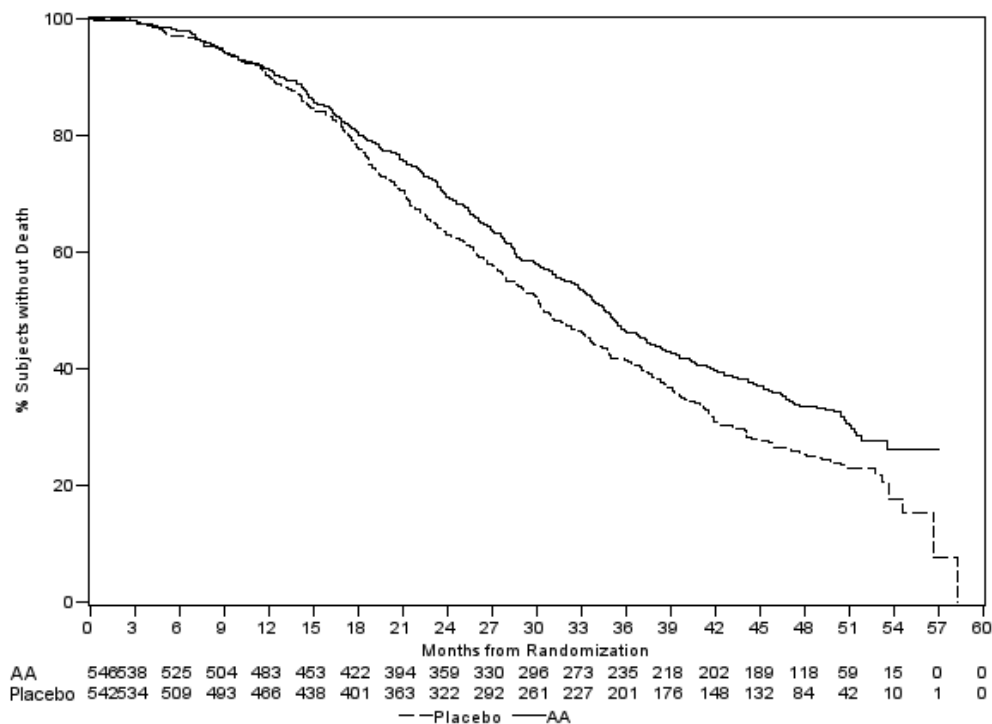
	ZYTIGA (N = 546)	PLACEBO (N = 542)
Interim Survival Analysis		
Deaths	147 (27%)	186 (34%)
Median overall survival in months (95% CI)	Not reached (NE, NE)	27.2 (25.95, NE)
p- value*	0.0097	
Hazard ratio** (95% CI)	0.752 (0.606, 0.934)	
Final Survival Analysis		
Deaths	354 (65%)	387 (71%)
Median overall survival in months (95% CI)	34.7 (32.7, 36.8)	30.3 (28.7, 33.3)
p-value*	0.0033	
Hazard ratio** (95% CI)	0.806 (0.697, 0.931)	

NE= Not Estimated

* p-value is derived from a log-rank test stratified by baseline ECOG score (0 or 1)

** Hazard ratio < 1 favors ZYTIGA

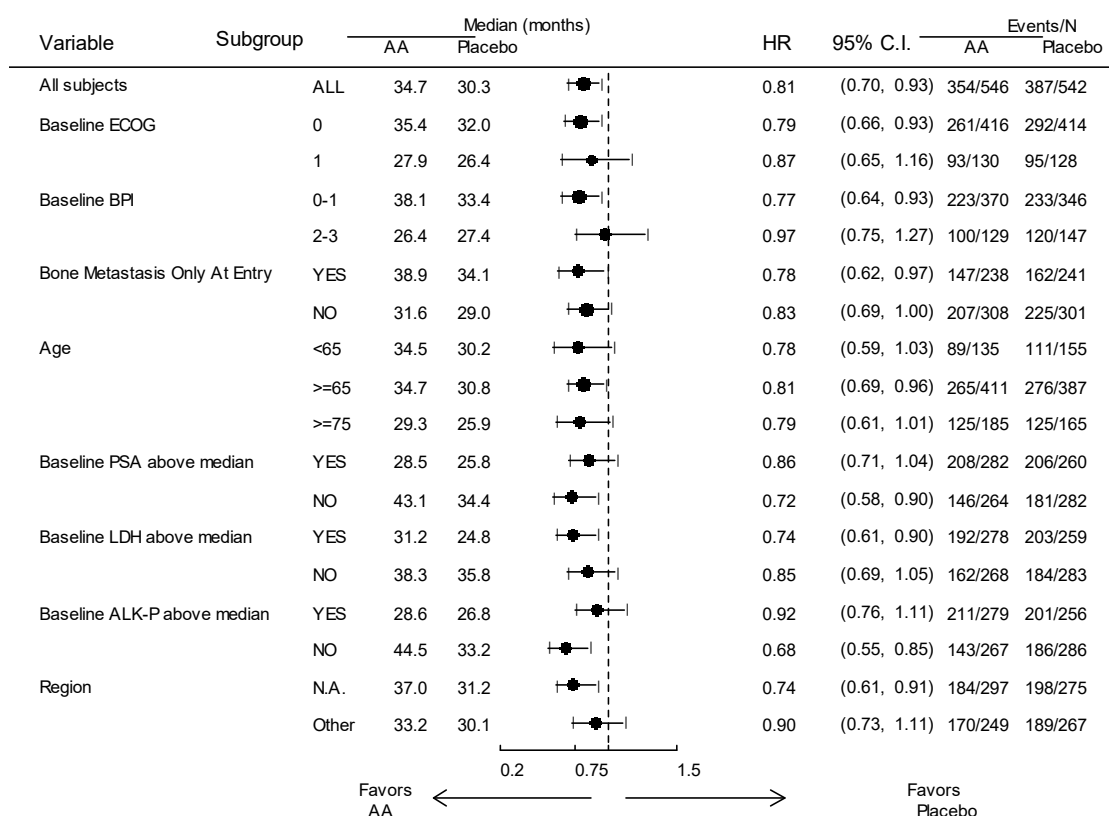
Figure 5: Kaplan Meier Survival Curves of Patients Treated with Either ZYTIGA or Placebo in Combination with Prednisone or Prednisolone plus LHRH Agonists or Prior Orchiectomy, Final analysis



AA = ZYTIGA

Subgroup analyses consistently favor treatment with ZYTIGA (see Figure 6).

Figure 6: Overall Survival by Subgroup: Hazard Ratio and 95% Confidence Interval



AA=ZYTIGA; ALK-P=alkaline phosphatase; BPI=Brief Pain Inventory; C.I.=confidence interval; ECOG=Eastern Cooperative Oncology Group performance score; HR=hazard ratio; LDH=lactic dehydrogenase; N.A.=North America; NE=not evaluable

In addition to the observed improvements in overall survival and rPFS, benefit was demonstrated for ZYTIGA vs. placebo treatment in all prospectively defined secondary endpoint measures as follows:

Time to PSA progression based on PCWG2 criteria: The median time to PSA progression was 11.1 months for patients receiving ZYTIGA and 5.6 months for patients receiving placebo (HR = 0.488; 95% CI: [0.420, 0.568], $p < 0.0001$). The time to PSA progression was approximately doubled with ZYTIGA treatment (HR = 0.488). The proportion of subjects with a confirmed PSA response was greater in the ZYTIGA group than in the placebo group (62% versus 24%; $p < 0.0001$).

Time to opiate use for cancer pain: The median time to opiate use for prostate cancer pain at the time of final analysis was 33.4 months for patients receiving ZYTIGA and was 23.4 months for patients receiving placebo (HR = 0.721; 95% CI: [0.614, 0.846], $p \leq 0.0001$).

Time to initiation of cytotoxic chemotherapy: The median time to initiation of cytotoxic chemotherapy was 25.2 months for patients receiving ZYTIGA and 16.8 months for patients receiving placebo (HR = 0.580; 95% CI: [0.487, 0.691], $p < 0.0001$).

Time to deterioration in ECOG performance score by ≥ 1 point: The median time to deterioration in ECOG performance score by ≥ 1 point was 12.3 months for patients receiving ZYTIGA and 10.9 months for patients receiving placebo (HR = 0.821; 95% CI: [0.714, 0.943], $p = 0.0053$).

The following study endpoints demonstrated a statistically significant advantage in favor of ZYTIGA treatment:

Objective response: Objective response was defined as the proportion of subjects with measurable disease achieving a complete or partial response according to RECIST criteria (baseline lymph node size was required to be ≥ 2 cm to be considered a target lesion). The proportion of subjects with measurable disease at baseline who had an objective response was 36% in the ZYTIGA group and 16% in the placebo group ($p < 0.0001$).

Pain: Treatment with ZYTIGA significantly reduced the risk of average pain intensity progression by 18% compared with placebo ($p=0.0490$). The median time to progression was 26.7 months in the ZYTIGA group and 18.4 months in the placebo group.

Time to degradation in the FACT-P (Total Score): Treatment with ZYTIGA decreased the risk of FACT-P (Total Score) degradation by 22% compared with placebo ($p = 0.0028$). The median time to degradation in FACT-P (Total Score) was 12.7 months in the ZYTIGA group and 8.3 months in the placebo group.

Study 301 (patients who had received prior chemotherapy)

Eleven percent of patients enrolled in Study 301 had an ECOG performance score of 2; 70% had radiographic evidence of disease progression with or without PSA progression; 70% had received one prior cytotoxic chemotherapy and 30% received two. Liver metastasis was present in 11% of patients treated with ZYTIGA.

It was recommended that patients be maintained on their study drugs until there was PSA progression (confirmed 25% increase over the patient's baseline/nadir) together with protocol-defined radiographic progression and symptomatic or clinical progression. The primary efficacy endpoint was overall survival.

In a planned analysis conducted after 552 deaths were observed, 42% (333 of 797) of patients treated with ZYTIGA, compared with 55% (219 of 398) of patients treated with placebo, had died. A statistically significant improvement in median overall survival was seen in patients treated with ZYTIGA (see Table 7 and Figure 7). An updated survival analysis was conducted when 775 deaths (97% of the planned number of deaths for the final analysis) were observed. Results from this updated survival analysis were consistent with those in the primary survival analysis (see Table 3).

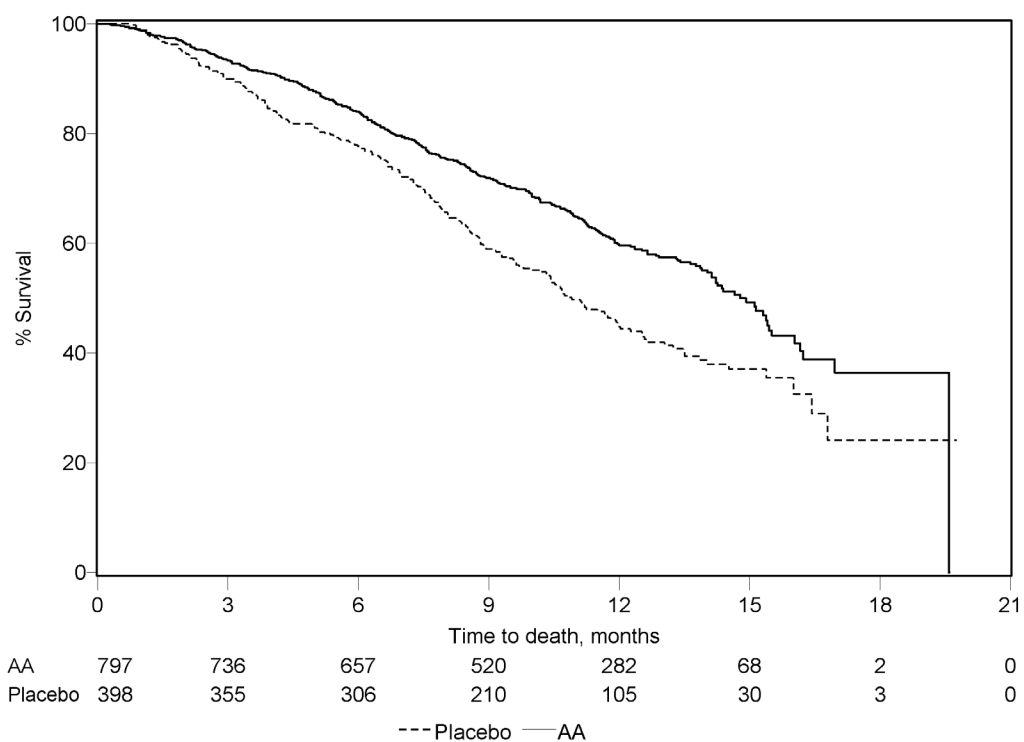
Table 7: Study 301: Overall Survival Of Patients Treated With Either ZYTIGA Or Placebo In Combination With Prednisone Or Prednisolone Plus LHRH Agonists Or Prior Orchiectomy

	ZYTIGA (N = 797)	PLACEBO (N = 398)
Primary Survival Analysis		
Deaths	333 (42%)	219 (55%)
Median overall survival in months (95% CI)	14.8 (14.1, 15.4)	10.9 (10.2, 12.0)
p-value	< 0.0001	
Hazard ratio* (95% CI)	0.646 (0.543, 0.768)	
Updated Survival Analysis		
Deaths	501 (63%)	274 (69%)
Median overall survival in months (95% CI)	15.8 (14.8, 17.0)	11.2 (10.4, 13.1)
Hazard ratio* (95% CI)	0.740 (0.638, 0.859)	

* Hazard ratio < 1 favors ZYTIGA

At all evaluation time points after the initial few months of treatment, a higher proportion of patients treated with ZYTIGA remained alive, compared with the proportion of patients treated with placebo (see Figure 7).

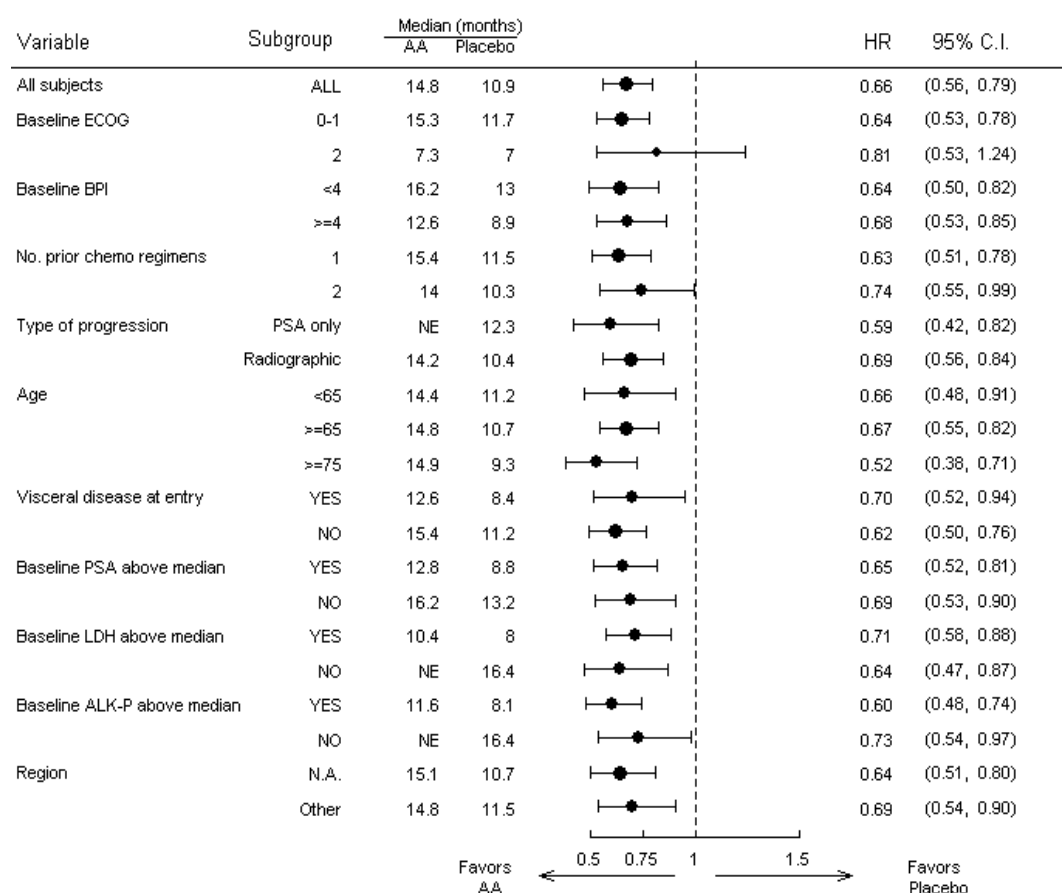
Figure 7: Kaplan Meier survival curves of patients treated with either ZYTIGA or placebo in combination with prednisone or prednisolone plus LHRH agonists or prior orchiectomy



AA = ZYTIGA

Subgroup survival analyses showed a consistent survival benefit for treatment with ZYTIGA (see Figure 8).

Figure 8: Overall Survival By Subgroup: Hazard Ratio And 95% Confidence Interval



AA=ZYTIGA; ALK-P=alkaline phosphatase; BPI=Brief Pain Inventory; C.I.=confidence interval; ECOG=Eastern Cooperative Oncology Group performance score; HR=hazard ratio; LDH=lactic dehydrogenase; N.A.= North America; NE=not evaluable

In addition to the observed improvement in overall survival, all secondary study endpoints favored ZYTIGA and were statistically significant after adjusting for multiple testing as follows:

Patients receiving ZYTIGA demonstrated a significantly higher total PSA response rate (defined as a ≥ 50% reduction from baseline), compared with patients receiving placebo, 38% versus 10%, $p < 0.0001$.

The median time to PSA progression was 10.2 months for patients treated with ZYTIGA and 6.6 months for patients treated with placebo (HR = 0.580; 95% CI: [0.462; 0.728], $p < 0.0001$).

The median radiographic progression-free survival was 5.6 months for patients treated with ZYTIGA and 3.6 months for patients who received placebo (HR = 0.673; 95% CI: [0.585; 0.776], $p < 0.0001$).

Pain

The proportion of patients with pain palliation was statistically significantly higher in the ZYTIGA group than in the placebo group (44% versus 27%, $p = 0.0002$). A responder for pain palliation was defined as a patient who experienced at least a 30% reduction from baseline in the BPI-SF worst pain intensity score over the last 24 hours without any increase in analgesic usage score observed at two consecutive evaluations four weeks apart. Only patients with a baseline pain score of ≥ 4 and at least one post-baseline pain score were analyzed ($n = 512$) for pain palliation.

A lower proportion of patients treated with ZYTIGA had pain progression compared to patients taking placebo at 6 (22% versus 28%), 12 (30% versus 38%) and 18 months (35% versus 46%). Pain progression was defined as an increase from baseline of $\geq 30\%$ in the BPI-SF worst pain intensity score over the previous 24 hours without a decrease in analgesic usage score observed at two consecutive visits, or an increase of $\geq 30\%$ in analgesic usage score observed at two consecutive visits. The time to pain progression at the 25th percentile was 7.4 months in the ZYTIGA group, versus 4.7 months in the placebo group.

Skeletal-Related events

A lower proportion of patients in the ZYTIGA group had skeletal-related events compared with the placebo group at 6 months (18% vs. 28%), 12 months (30% vs. 40%), and 18 months (35% vs. 40%). The time to first skeletal-related event at the 25th percentile in the ZYTIGA group was twice that of the control group at 9.9 months vs. 4.9 months. A skeletal-related event was defined as a pathological fracture, spinal cord compression, palliative radiation to bone, or surgery to bone.

5.2 Pharmacokinetic Properties

General introduction

Following administration of abiraterone acetate, the pharmacokinetics of abiraterone has been studied in healthy subjects, patients with metastatic advanced prostate cancer and subjects without cancer with hepatic or renal impairment. Abiraterone acetate is rapidly converted *in vivo* to abiraterone, an androgen biosynthesis inhibitor (see 5.1-Mechanism of action).

Absorption

Following oral administration of abiraterone acetate in the fasting state, the time to reach maximum plasma abiraterone concentration is approximately 2 hours.

Administration of abiraterone acetate with food, compared with administration in a fasted state, results in up to a 17-fold increase in mean systemic exposure of abiraterone, depending on the fat content of the meal. Given the normal variation in the content and composition of meals, taking ZYTIGA with meals has the potential to result in highly variable exposures. Therefore, **ZYTIGA must not be taken with food**. ZYTIGA tablets must be taken as a single dose once daily on an empty stomach. ZYTIGA must be taken at least two hours after eating and food must not be eaten for at least one hour after taking ZYTIGA. The tablets must be swallowed whole with water (see 4.2).

Distribution and protein binding

The plasma protein binding of ^{14}C -abiraterone in human plasma is 99.8%. The apparent volume of distribution is approximately 5630 L, suggesting that abiraterone extensively distributes to peripheral tissues.

Metabolism

Following oral administration of ^{14}C -abiraterone acetate as capsules, abiraterone acetate is hydrolyzed to abiraterone, which then undergoes metabolism including sulphation, hydroxylation and oxidation primarily in the liver. The majority of circulating radioactivity (approximately 92%) is found in the form of metabolites of abiraterone. Of 15 detectable metabolites, 2 main metabolites, abiraterone sulphate and N-oxide abiraterone sulphate, each represents approximately 43% of total radioactivity.

Elimination

The mean half-life of abiraterone in plasma is approximately 15 hours based on data from healthy subjects. Following oral administration of ^{14}C -abiraterone acetate, approximately 88% of the radioactive dose is recovered in feces and approximately 5% in urine. The major compounds present in feces are unchanged abiraterone acetate and abiraterone (approximately 55% and 22% of the administered dose, respectively).

Special populations

Renal impairment

The pharmacokinetics of abiraterone was compared in patients with end-stage renal disease on a stable hemodialysis schedule, versus matched control subjects with normal renal function. Systemic exposure to abiraterone after a single oral 1000 mg dose did not increase in patients with end-stage renal disease on dialysis.

Administration of ZYTIGA in patients with renal impairment including severe renal impairment does not require dose reduction (see 4.2).

Hepatic impairment

The pharmacokinetics of abiraterone acetate was examined in subjects with pre-existing mild or moderate hepatic impairment (Child-Pugh Class A and B, respectively) and in healthy control subjects. Systemic exposure to abiraterone after a single oral 1,000 mg dose increased by approximately 11% and 260% in subjects with mild and moderate pre-existing hepatic impairment, respectively. The mean half-life of abiraterone is prolonged to approximately 18 hours in subjects with mild hepatic impairment and to approximately 19 hours in subjects with moderate hepatic impairment. No dose adjustment is necessary for patients with pre-existing mild hepatic impairment. There are no data on the clinical safety and efficacy of multiple doses of abiraterone acetate when administered to patients with moderate or severe hepatic impairment (Child Pugh Class B or C). No dosage adjustment can be predicted. ZYTIGA should be used with caution in patients with moderate hepatic impairment, only if the benefit clearly outweighs the possible risk (see 4.2 and 4.4-Hepatotoxicity and Hepatic impairment). ZYTIGA should not be used in patients with severe hepatic impairment. For patients who develop hepatotoxicity during treatment with ZYTIGA, suspension of treatment and dosage adjustment may be required (see 4.2 and 4.4- Hepatotoxicity and Hepatic impairment).

Effects on the QT interval

In a cardiovascular safety study in patients with metastatic advanced prostate cancer there were no significant effects of abiraterone acetate on the cardiac QT/QTc interval.

5.3 Non-Clinical Information

Carcinogenicity and mutagenicity

Abiraterone acetate was not carcinogenic in a 6-month study in the transgenic (Tg.rasH2) mouse. In a 24-month carcinogenicity study in the rat, abiraterone acetate increased the incidence of interstitial cell neoplasms in the testes. This finding is considered related to the pharmacological action of abiraterone and rat specific. Abiraterone acetate was not carcinogenic in female rats.

Abiraterone acetate and abiraterone were devoid of genotoxic potential in the standard panel of genotoxicity tests, including an in vitro bacterial reverse mutation assay (the Ames test), an in vitro mammalian chromosome aberration test (using human lymphocytes) and an in vivo rat micronucleus assay.

Reproductive toxicology

In fertility studies in both male and female rats, abiraterone acetate reduced fertility, which was completely reversible in 4 to 16 weeks after abiraterone acetate was stopped.

In a developmental toxicity study in the rat, abiraterone acetate affected pregnancy including reduced fetal weight and survival. Effects on the external genitalia were observed though abiraterone acetate was not teratogenic.

In these fertility and developmental toxicity studies performed in the rat, all effects were related to the pharmacological activity of abiraterone.

ZYTIGA is contraindicated in pregnancy (see 4.3 and 4.6-Pregnancy).

Animal toxicology

In all animal toxicity studies, circulating testosterone levels were significantly reduced. As a result, reduction in organ weights and morphological and/or histopathological changes in the reproductive organs, and the adrenal, pituitary and mammary glands were observed. All changes showed complete or partial reversibility. The changes in the reproductive organs and androgen-sensitive organs are consistent with the pharmacology of abiraterone. All treatment-related hormonal changes reversed or were shown to be resolving after a 4-week recovery period.

6. PHARMACEUTICAL PARTICULARS

6.1 List of Excipients

ZYTIGA 250 mg uncoated tablets contain the following excipients:

colloidal silicon dioxide
croscarmellose sodium
lactose monohydrate
magnesium stearate
microcrystalline cellulose
povidone
sodium lauryl sulfate

ZYTIGA 500 mg film-coated tablets contain the following excipients:

Tablet core:

colloidal silicon dioxide
croscarmellose sodium
hypromellose
lactose monohydrate
magnesium stearate
silicified microcrystalline cellulose
sodium lauryl sulfate

Tablet film-coat:

iron oxide black
iron oxide red
macrogol 3350
polyvinyl alcohol
talc
titanium dioxide

6.2 Incompatibilities

Not applicable.

6.3 Shelf life

ZYTIGA 250 mg tablets

24 months

See expiry date on the outer pack.

ZYTIGA 500 mg film-coated tablets

36 months

See expiry date on the outer pack

6.4 Storage Conditions

Store below 30°C.

Keep out of sight and reach of children.

6.5 Nature and Contents of Container

ZYTIGA 250 mg tablets

ZYTIGA is available in high-density polyethylene round white bottles fitted with a polypropylene cap. Package size is 120 tablets.

ZYTIGA 500 mg film-coated tablets

ZYTIGA 500 mg tablets are available in blister packs for a total of 60 tablets. Individual (PVDC/alu) blister strips are packed inside a folding carton.

6.6 Instructions for Use and Handling and Disposal

Based on its mechanism of action, ZYTIGA may harm a developing fetus; therefore, women who are pregnant or women who may be pregnant should not handle ZYTIGA 250 mg uncoated tablets without protection, e.g., gloves (see 4.6-Pregnancy).

Any unused product or waste material should be disposed of in accordance with local requirements.

7. MANUFACTURER

ZYTIGA 250 mg tablets

Patheon Inc.,
2100 Syntex Court,
Mississauga, Ontario,
L5N 7K9, Canada.

ZYTIGA 500 mg film-coated tablets

Patheon France S.A.S
40 Boulevard de Champaret,
38300 Bourgoin Jallieu,
France.

8. PRODUCT REGISTRATION HOLDER

Johnson & Johnson Sdn Bhd (3718-D)
Level 8, The Pinnacle,
Persiaran Lagoon, Bandar Sunway,
46150, Petaling Jaya, Selangor, Malaysia

9. DATE OF REVISION OF THE TEXT

19 September 2022 (CCDS v19Mar2021)
